Exercise: Enhanced Cashier System with Functions

Objective:

Design a Cashier System that:
1. Allows multiple product details to be added to a bill.
2. Applies a membership discount, if applicable.
3. Applies a voucher discount, if applicable, up to a maximum of 5%.
4. Calculates and displays the final bill with all items, discounts, and total amount due.

This project will give students hands-on practice with user-defined functions, selection control structures (using if-else for discounts), repetition control structures (using loops for multiple products), along with concepts like data types, operators, and input/output.

Instructions:

Required User-Defined Functions

1. `add_product`

A function that prompts the user to enter the product name, unit price, and quantity purchased.
Calculates the total price for this product with any applicable bulk discount:
- 10% discount if the quantity is greater than 10 units.
- 5% discount if the quantity is between 5 and 10 units.
- No discount if the quantity is fewer than 5 units.
Returns the discounted total for the product.

2. `apply_membership_discount`

A function that asks the user if the customer has a membership.
If yes, apply an additional 2.5% membership discount on the total bill.

3. `apply_voucher_discount`

A function that prompts the user to enter a voucher discount.
Limits the voucher discount to a maximum of 5%.

4. `display_final_bill`

A function that displays the names, unit prices, quantities, and discounted prices for each item.
Calculates and displays the grand total amount due, with all discounts applied.

Program Flow

1. Display Menu Options

Start the program by displaying a menu with the following options:
- 1. Add Product to Bill
- 2. Apply Membership Discount
- 3. Apply Voucher Discount
- 4. Display Final Bill and Exit

2. Add Product to Bill

For each product, use `add_product` to gather details, calculate, and store the total price for the final bill.

3. Apply Membership Discount

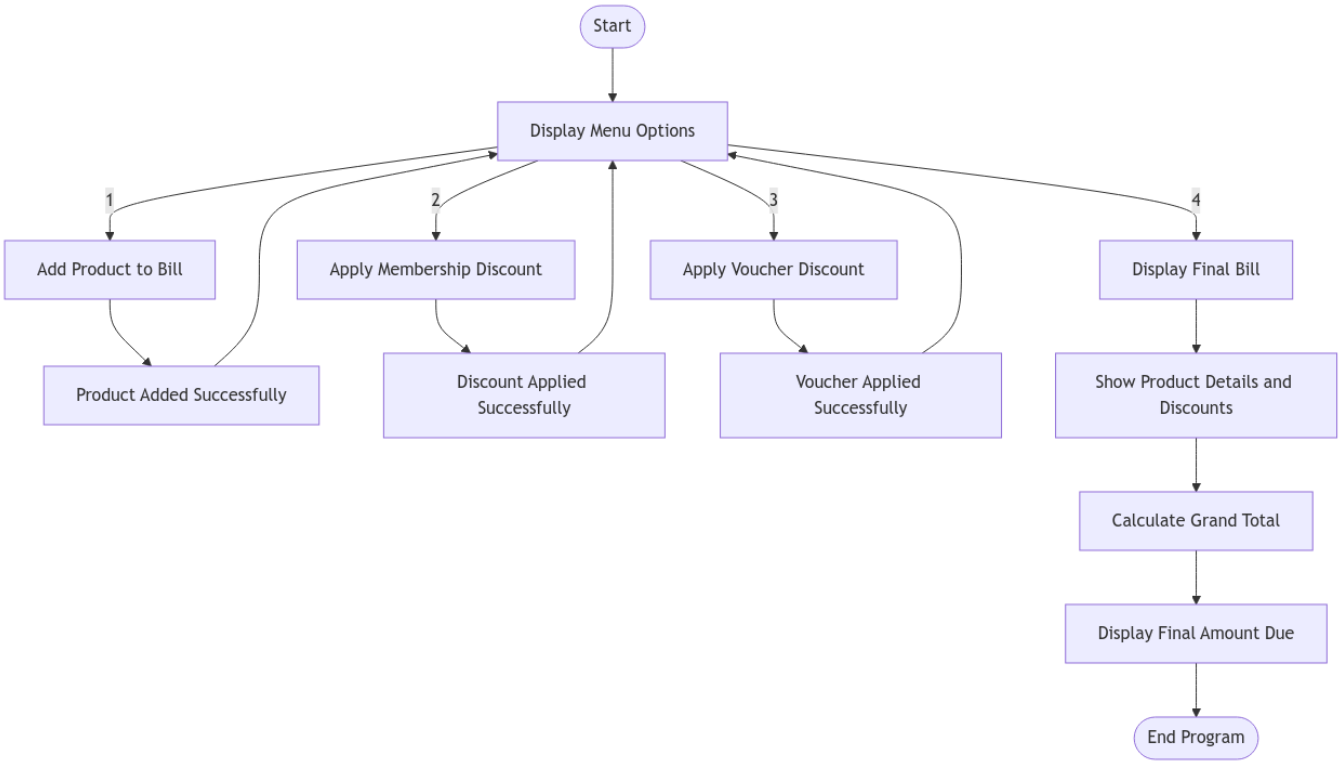
Use `apply_membership_discount` to check for membership and apply the discount.

4. Apply Voucher Discount

Use `apply_voucher_discount` to add a voucher discount (up to 5%).

5. Display Final Bill and Exit

Use `display_final_bill` to show all stored product details with the grand total at the end.



Sample Terminal Output:

```
Cashier System
1. Add Product to Bill
2. Apply Membership Discount
3. Apply Voucher Discount
4. Display Final Bill and Exit
Enter your choice: 1

Enter Product Name: Oranges
Enter Unit Price: 3.0
Enter Quantity Purchased: 8
Product added with a 5% bulk discount.

Cashier System
1. Add Product to Bill
2. Apply Membership Discount
3. Apply Voucher Discount
4. Display Final Bill and Exit
Enter your choice: 2

Does the customer have a membership? (yes/no): yes
2.5% membership discount applied.
```

```
Cashier System
1. Add Product to Bill
2. Apply Membership Discount
3. Apply Voucher Discount
4. Display Final Bill and Exit
Enter your choice: 3

Enter voucher discount (max 5%): 4.5%
Voucher discount applied.

Cashier System
1. Add Product to Bill
2. Apply Membership Discount
3. Apply Voucher Discount
4. Display Final Bill and Exit
Enter your choice: 4

Final Bill:
Product    | Unit Price | Quantity | Total Cost (Discount Applied)
-----------------------------------------------------------------
Oranges    | $3.00      | 8        | $22.80
-----------------------------------------------------------------
Membership Discount: 2.5%
Voucher Discount: 4.5%
-----------------------------------------------------------------
Grand Total Amount Due: $21.59
```